

TinySea

Simulation Diagrams

Headless ecosystem simulator
Visual reference

Model version: v12-per-species-tracking
Source: tinysea/Assets/scripts/Simulation/
Generated: 2026-05-01

Northeastern University GhostLab

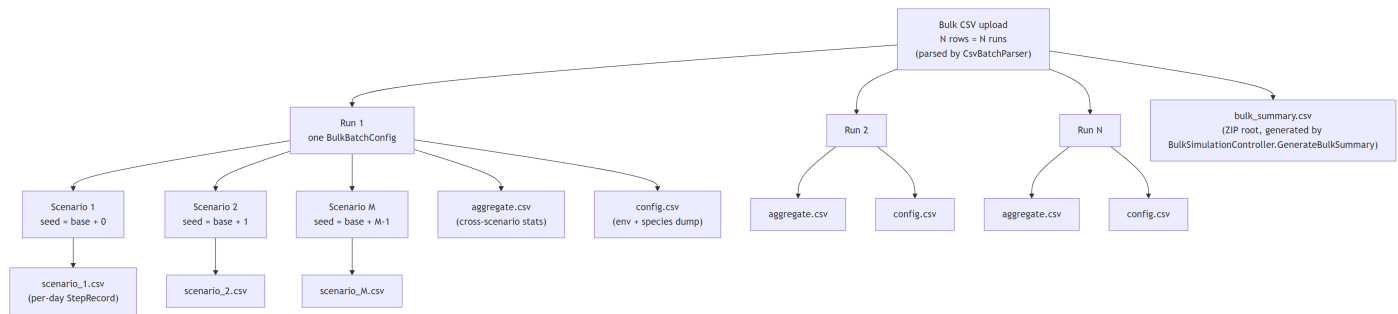
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10	Fractional-event accumulator pattern	How births / predation / natural death / condition death track fractional events across days: rawEvents -> accumulator[FullName] -> floor() -> integer apply, residual carries to next day.

One Mermaid diagram per page. Each is rendered from the source-controlled **.md** file under `tinysea/docs/diagrams/`. If a diagram and the source code disagree, trust the source and update the diagram.

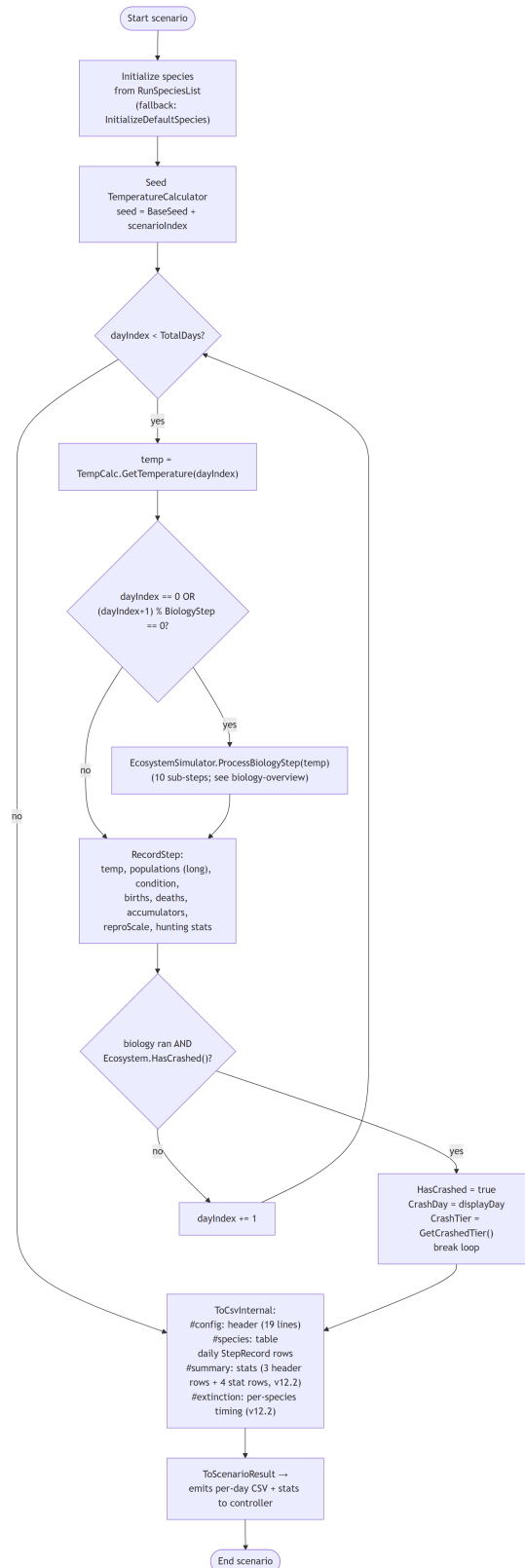
1. Bulk → Run → Scenario hierarchy

How a bulk CSV upload becomes N runs, each producing M scenarios and output files. Source: BulkSimulationController.cs, SimulationController.cs, CsvBatchParser.cs.



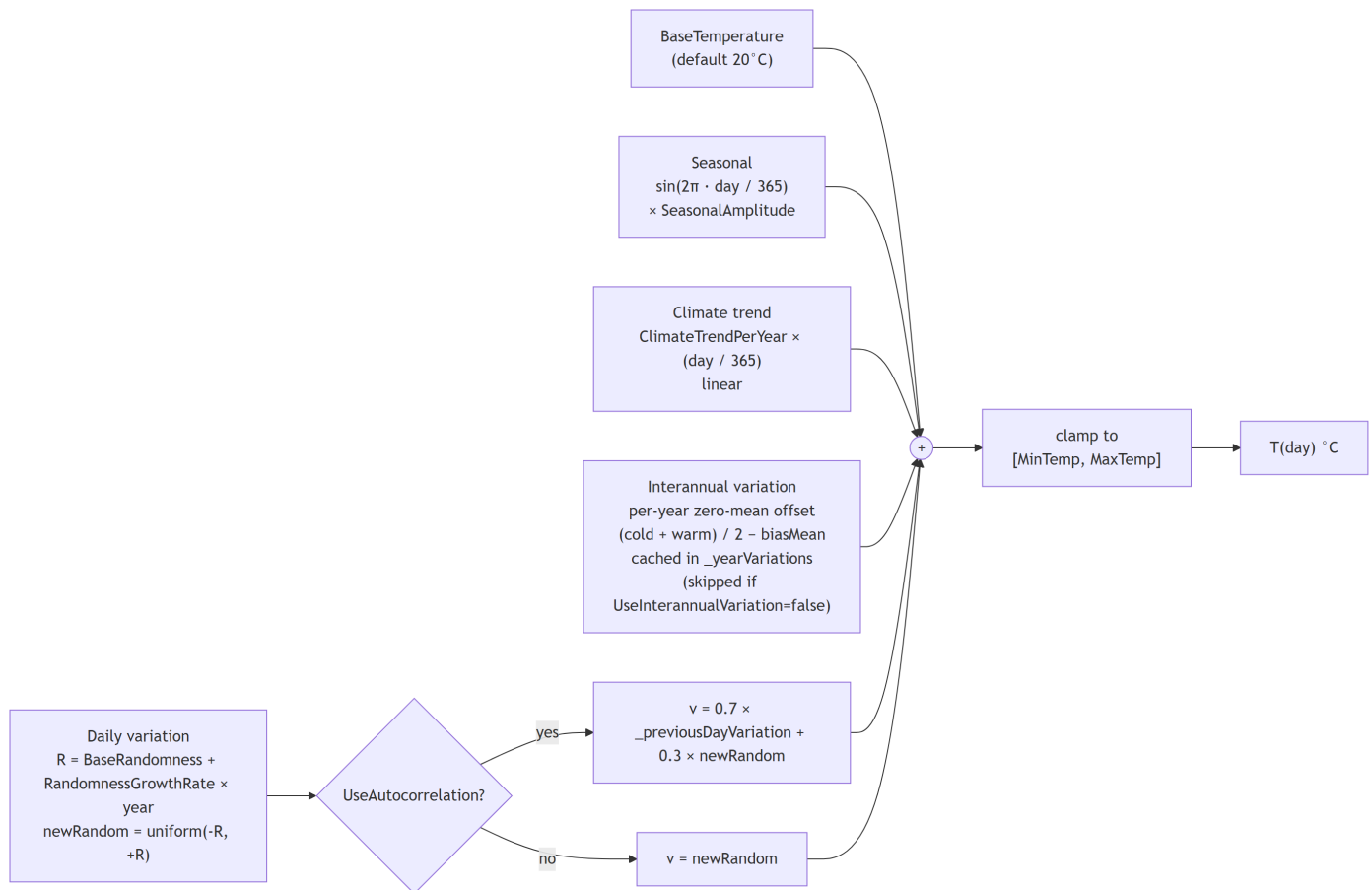
2. Scenario flow

Per-scenario day loop: temperature → biology dispatch → record StepRecord → crash check → CSV emission. Source: SimulationRunner.Run, EcosystemSimulator.HasCrashed.



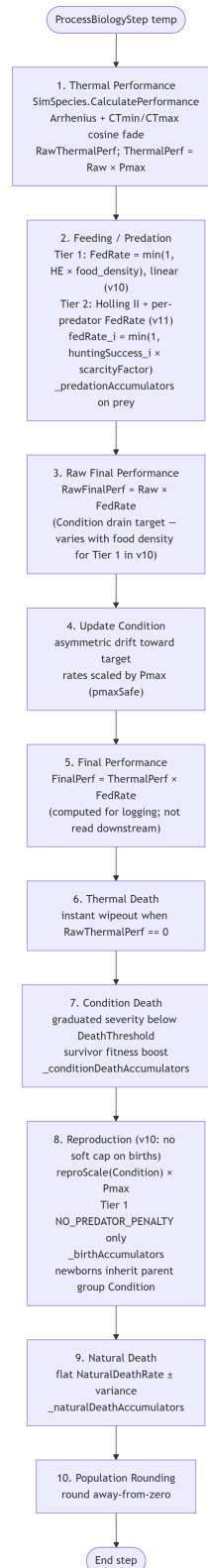
3. Temperature model

Daily temperature = BaseTemperature + seasonal sinusoid + linear climate trend + interannual variation (zero-mean, biasMean-corrected) + autocorrelated daily noise, clamped to [MinTemp, MaxTemp]. Source: TemperatureCalculator.cs.



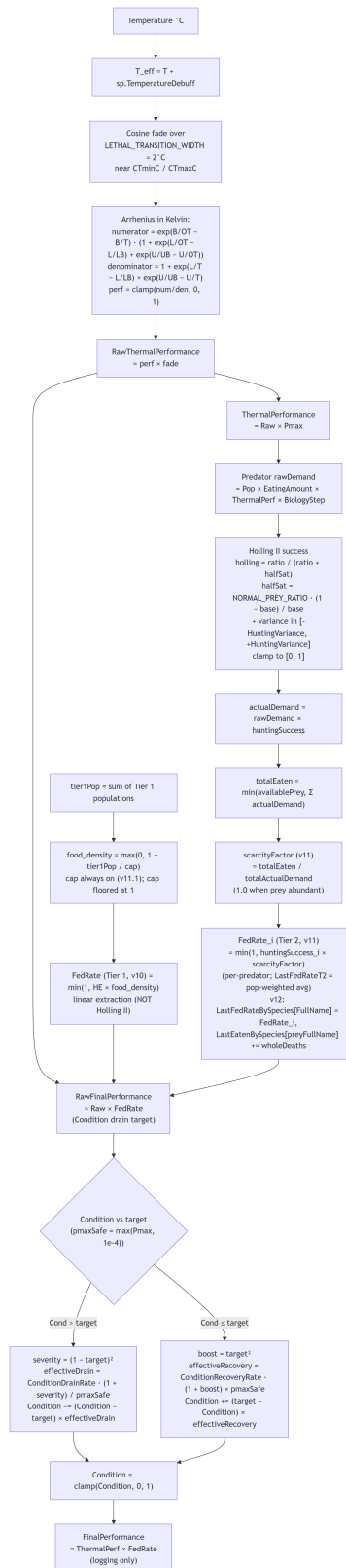
4. Biology — 10-step overview

The 10 sub-steps invoked each day in `EcosystemSimulator.ProcessBiologyStep`: thermal performance → feeding/predation → raw final performance → update Condition → final performance → thermal death → condition death → reproduction → natural death → population rounding.



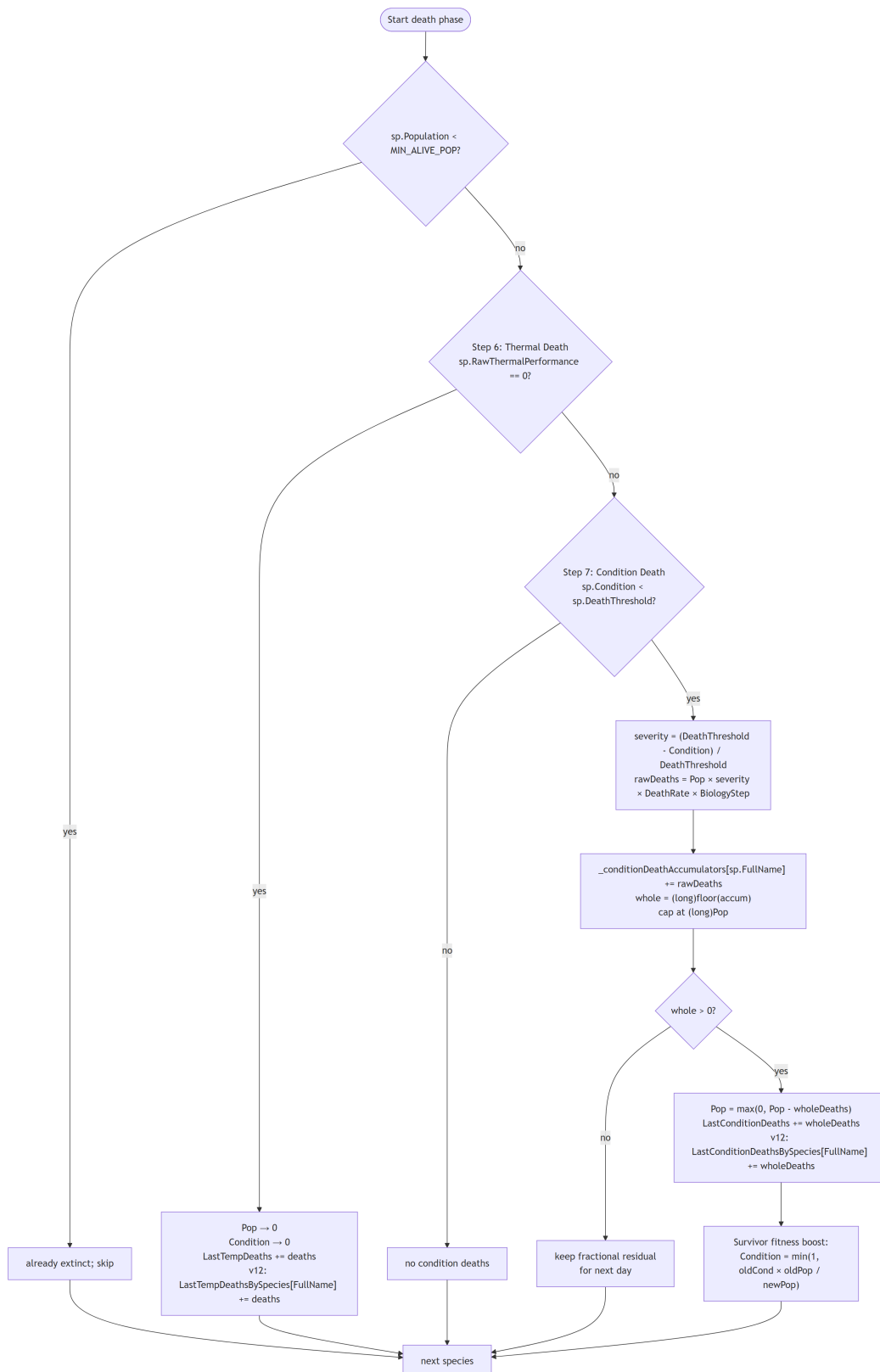
5. Biology phase 1: performance (Steps 1–5)

Arrhenius thermal performance with cosine fade near CTmin/CTmax, Tier 1 food-pool linear FedRate (v10), Tier 2 Holling II + per-predator FedRate (v11), Condition update with Pmax rate scaling (v9). Source: SimSpecies.CalculatePerformance, EcosystemSimulator.ProcessFeedingWithAccumulator, UpdateCondition.



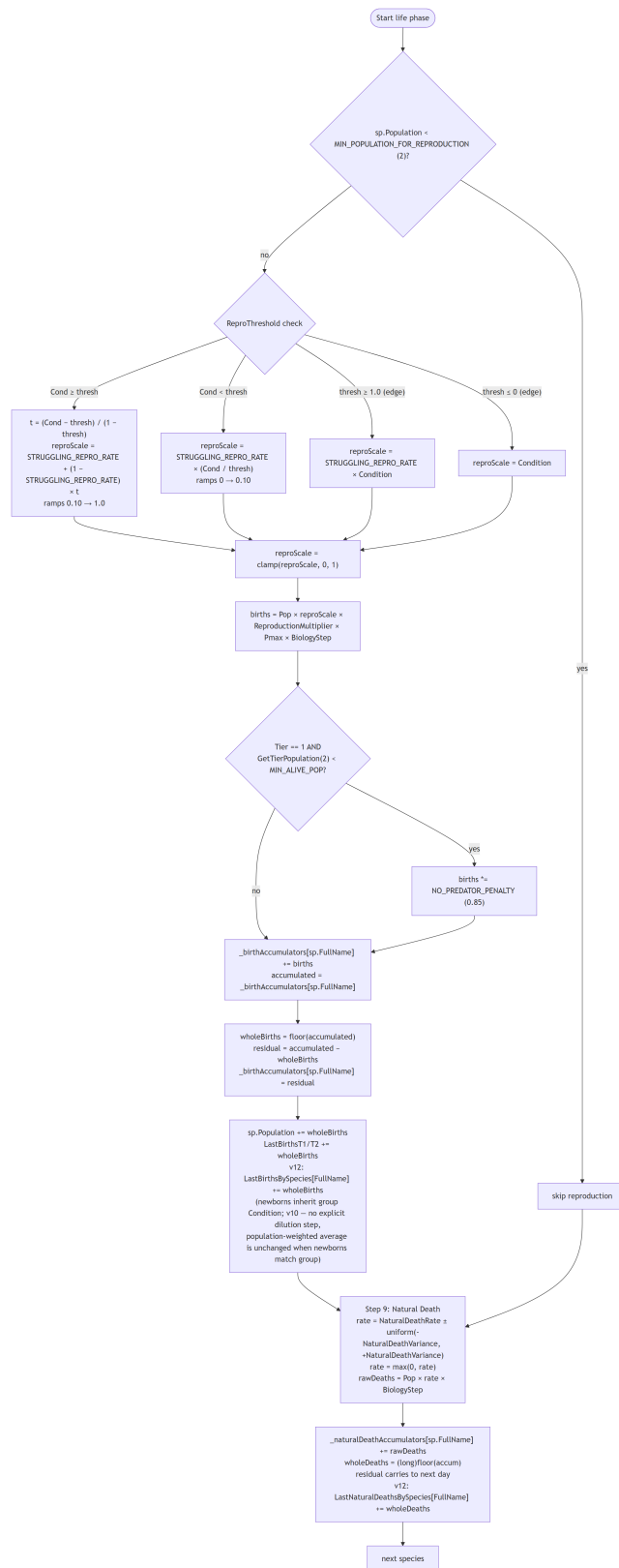
6. Biology phase 2: death (Steps 6–7)

Thermal death (instant wipeout when $\text{RawThermalPerformance} == 0$) and condition death (graduated severity below $\text{DeathThreshold} + \text{survivor fitness boost to prevent death spirals}$). Source: `EcosystemSimulator.ApplyThermalDeath`, `ApplyConditionDeath`.



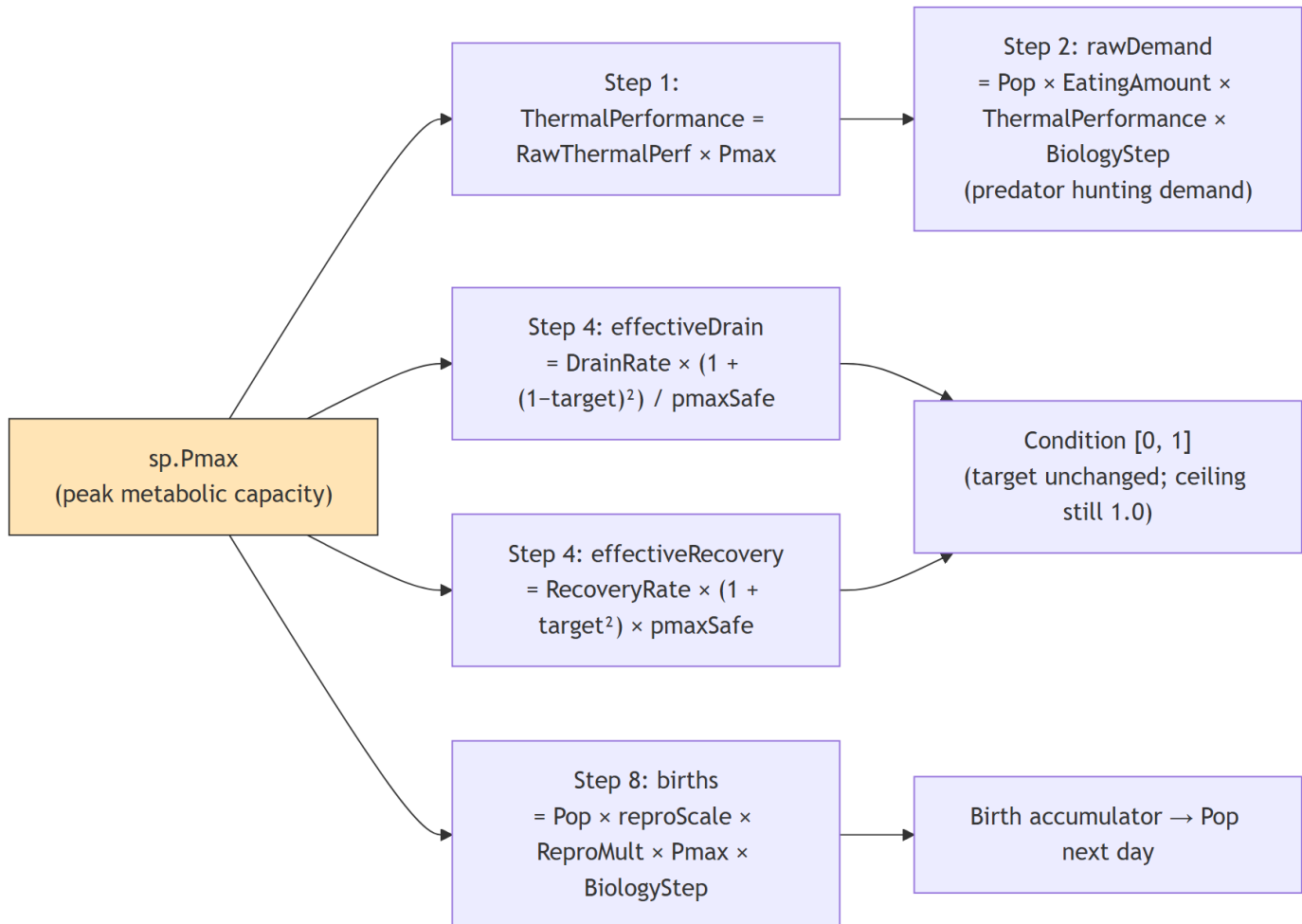
7. Biology phase 3: life (Steps 8–9)

Reproduction with piecewise reproScale joined at STRUGGLING_REPRO_RATE = 0.10, Pmax birth multiplier (v9), Tier 1 NO_PREDATOR_PENALTY = 0.85, fractional birth accumulator. Newborns inherit parent group Condition (v10). Natural death is flat-rate, performance-independent. Source: EcosystemSimulator.ApplyReproduction, ApplyNaturalDeathWithAccumulator.



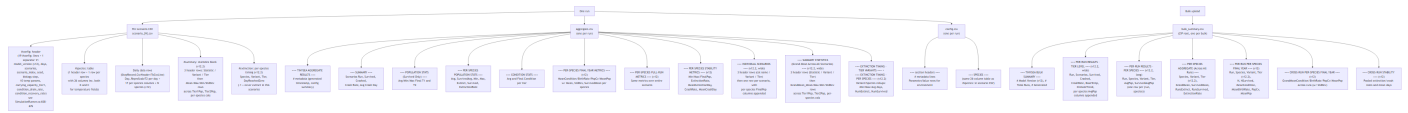
8. Pmax flow through the pipeline

The four places Pmax enters biology, all post-Condition: Step 1 thermal performance scaling, Step 4 Condition drain divisor, Step 4 Condition recovery multiplier, Step 8 birth multiplier. Pmax is deliberately absent from the Condition target so thresholds are species-agnostic.



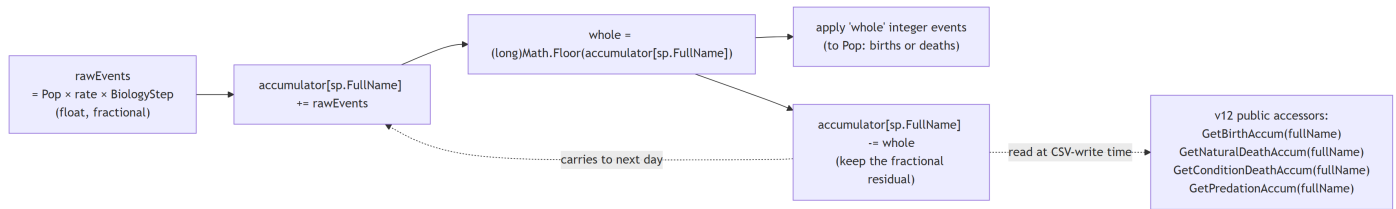
9. CSV output shape

Per-scenario CSV (#config: header + #species: table + 41 daily columns + 17 v12 per-species columns + #summary + #extinction), aggregate.csv (10+ === SECTION === blocks including v12 per-species metrics), and bulk_summary.csv (6 sections, with v12.3 cross-run filtering for Condition). Source: `SimulationRunner.ToCsvInternal`, `ScenarioResult.ToAggregateCsv`, `BulkSimulationController.GenerateBulkSummary`.



10. Fractional-event accumulator pattern

How births / predation / natural death / condition death track fractional events across days: rawEvents -> accumulator[FullName] -> floor() -> integer apply, residual carries to next day. v12 adds public accessors GetBirthAccum / GetNaturalDeathAccum / GetConditionDeathAccum / GetPredationAccum. `_thermalDeathAccumulators` is declared but unused (thermal death is binary).



Sources of truth

Area	File
Biology sequence	Assets/scripts/Simulation/EcosystemSimulator.cs
Per-species fields & Arrhenius	Assets/scripts/Simulation/SimSpecies.cs
Temperature model	Assets/scripts/Simulation/TemperatureCalculator.cs
Scenario orchestration & CSV	Assets/scripts/Simulation/SimulationRunner.cs
Per-scenario / aggregate stats	Assets/scripts/Simulation/DataStructure/ScenarioResult.cs
Bulk orchestration & CSV	Assets/scripts/Simulation/BulkSimulationController.cs
Bulk row schema	Assets/scripts/Simulation/BulkBatchConfig.cs
Bulk CSV parser	Assets/scripts/Simulation/CsvBatchParser.cs
Config schema	Assets/scripts/Simulation/DataStructure/SimulationConfig.cs

See [docs/simulation-spec.md](#) and [docs/csv-formats.md](#) for the textual spec; [docs/pending-list.md](#) for known issues.